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Deriving Design Feature Vectors for Patent Images Using Convolutional Neural Networks

The patent database is often used by designers to search for inspirational stimuli for innovative design opportunities because of the large size, extensive variety, and the massive quantity of design information contained in patent documents. Growing work on design-by-analogy has adopted various vectorization approaches for associating design documents. However, they only focused on text analysis and ignored visual information. Research in engineering design and cognitive psychology has shown that visual stimuli may benefit design ideation. In this study, we focus on visual design stimuli and automatically derive the vector space and the design feature vectors representing design images. The automatic vectorization approach uses a novel convolutional neural network architecture named Dual-Visual Geometry Group (VGG) aiming to accomplish two tasks: visual material-type prediction and international patent classification (IPC) section-label predictions. The derived feature vectors that embed both visual characteristics and technology-related knowledge can be potentially utilized to guide the retrieval and use of near-field and far-field design stimuli according to their vector distances. We report the accuracy of the training tasks and also use a case study to demonstrate the advantages of design image retrievals based on our model. [DOI: 10.1115/1.4049214]

Keywords: design-by-analogy, data-driven design, design representation, patent image, machine learning, convolutional neural network, artificial intelligence, computer-aided design, design automation

1 Introduction

Engineering designers often reuse and transform previous technical knowledge and ideas into their new designs [1–3]. In particular, previous research on design-by-analogy has shown that designers can gain inspiration from stimuli from both the same and different domains as the design problem and generate ideas in varied quality, quantity, and novelty [4–11]. Among the different kinds of data sources for design stimuli, the patent database is probably the biggest digitized design repository. Growing work on design-by-analogy has explored the patent database and proposed various vectorization approaches for associating patent documents as design stimuli of different analogical distances to given design problems [12–14]. However, most of these prior works only focused on text analysis and ignored visual information.

Prior studies have suggested the advantages of using visual information (design drawings) in problem-solving, e.g., providing perceptual inferences and enabling easy recognition of elements, such as spatial relations and shapes [15]. For instance, Ullman et al. studied the importance of drawings in the engineering design process [16]. In a patent, graphical information plays an important role in presenting its novelty. New technologies and innovations are often described with rapidly changing and inconsistent terms, but drawings can more effectively convey the same characteristics [17]. Some of the literature on design practices encourages collecting and seeking visual examples to inform the design process [18–23]. Some prior works have mined and analyzed patent images, but they only use geometric and local features to represent images without considering technology-related knowledge [24–28].

Herein, we aim at deriving the vector space and the vectors representing design images from the patent database, by considering both visual features and technology-related knowledge, to support the retrieval of visual design stimuli for design-by-analogy. This is not a trivial task, and it has several challenges. First, the patent database is vast. For example, the United States Patent and Trademark Office (USPTO) patent database alone contains more than ten million patent records. Each patent contains more than ten images on average. Second, all the images found in the patent files are binary (black and white) with no color information. Furthermore, the knowledge within the patent images is hard to represent due to their texture-less characteristics.

The current image representing and searching methods are mostly based on content-based image retrieval (CBIR) techniques, which exploit the color, shape, and textural information of images [29,30]. Some researchers use local features, such as scale-invariant feature transform (SIFT) features, to represent a patent image [17], and then use these features to search for similar images. The retrieval results only utilize the geometric information of the query image, ignoring technical-related knowledge. Compared to these

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local feature-based methods, "representation learning" that automatically learns features from the data is a recent and rapidly developing machine learning approach [31]. The technique used in this paper is a convolutional neural network (CNN), which is currently one of the most popular representation learning methods. CNN enables the learning of richer image features because of its deep architecture [32]. To the best of our knowledge, this is the first attempt using CNN to derive design feature vectors for patent images.

Specifically, we introduce a CNN-based approach for patent image vectorization, which may further support visual design stimuli retrieval to aid in design-by-analogy. In order for the CNN training to embed not only visual information but also technology-related knowledge into feature vectors, we propose a novel CNN architecture named Dual-VGG. It is trained with two different tasks: visual material type prediction and IPC section-label prediction. Specifically, visual material-type prediction is to classify patent images into nine predefined-type categories [25], namely, abstract drawing, flowchart, graph, chemical structure, table, DNA, math formula, computer program, and symbol. The first task enables our model to learn the visual characteristics of the patent images. In the IPC, the technical topics are categorized based on the knowledge of structure and function of the artifacts and are supplemented by their applications [33]. In this way, the IPC labels not only are the index of different groups but also contain technical-related knowledge of the structures, functions, mechanisms, and principles that might be useful for the design inspiration process [33]. Thus, the second task, the IPC section-label prediction, enables our model to learn technical-related knowledge.

After the training and validating process, the design features can be derived from the final second-to-last fully connected layer of the neural network. Cosine similarities are used as the analogical distance between the design feature vector, representing a given query image, and other images in the image repository. In turn, such distance information may guide the search and retrieval of patent images as visual design stimuli for design-by-analogy. Meanwhile, according to design creativity theories [5–11,34–37], near stimuli are beneficial for effectiveness and quality of design ideation, and far stimuli give rise to idea novelty and chances for breakthrough innovation. The work presented in this paper, by focusing on patent images and automatic learning of their design vector presentations with a novel CNN architecture, provides a new way to identify and determine near- versus far visual stimuli.

Therefore, this research contributes to the growing studies about data-driven design [1,38,39], design-by-analogy [8,9,13,40], and machine learning-based design analytics [41–45].

2 Literature Review

This paper builds on prior studies of data-driven design and design-by-analogy to develop a CNN-based design image vectorization and retrieval method to support engineering design ideation and exploration. In this section, we review the relevant literature on data-driven design, design-by-analogy, and patent image representation and retrieval.

2.1 Data-Driven Design. Several studies in the engineering design field have proposed big data techniques to aid design ideation and concept generation in the early design phase for designers. For example, design-by-analogy aims to extract functional analogies from general data sources to help designers in systematically searching and identifying relevant cases or knowledge [13,40]. The theory of inventive problem-solving (TRIZ) is a problem-solving and analysis tool developed from the study of patterns in the global patent documents [46]. The infused design method uses the metalevel representation of design problems to facilitate the discovery and use of knowledge across different technological fields [47]. Reich and Shai [48] proposed the engineering knowledge genome to enhance searching for common knowledge and method structures in interdisciplinary domains.

A few data-driven computer tools have been developed to implement these design methodologies. Chakrabarti et al. [49] developed the Idea-Inspire system for designers to retrieve descriptions of systems from artificial and natural fields that could act as stimuli for design ideation to solve a problem. Goel et al. [50] developed the design-by-analogy to nature engine (DANE) to identify specific kinds of analogical design activities and to facilitate studies of the cognitive foundations of analogical design. AskNature is a web-based tool [51], with more than 1,600 biological strategy cases in its repository, to provide biological inspirations to designers. Several recent studies have shown crowdsourced design idea data from online platforms, such as Amazon Mechanical Turk, can be mined and represented with machine learning techniques to be used as design inspiration stimuli [8,52].

Specifically, as the largest digitized design repository, the patent database is widely used to support design using various data mining techniques. For example, Cascini and Russo [53] developed a system named Pat-Analyzer to automatically identify the design contradictions contained in a patented invention in TRIZ by analyzing patent textual information. Mukherjea et al. [54] created a biomedical patent semantic web that can recommend patents based on the semantic relationship between biological terms from the patent text. Ji et al. [33] proposed a design inspiration method that uses the knowledge from the international patent classification of US patents for design education. Luo et al. [11,39,55] developed a computer-aided ideation system called InnoGPS using a network map of all patent classes to augment the exploration of design opportunities in the total technology space and to guide the retrieval of prior knowledge and concepts across domains for design analogies and syntheses. Sarica et al. [41] developed a semantic network of more than 4 million technical terms named TechNet based on the USPTO database for engineering knowledge discovery and ideation support. TechNet has been employed to augment design retrieval and exploration [56], patent document search [57], and idea generation and evaluation [58], among other applications.

However, these prior methods aimed to mine bibliographic and classification information of patents as well as patent textual data, ignoring image information. We should notice that many technological specifications are more appropriately described by technical figures in many domains, especially *mechanical design*. Thus, retrieving design information in images of a patent document can play a crucial role and enhance patent analytics for data-driven design and inspire design-by-analogy.

2.2 Design-by-Analogy and Analogical Distance. The process of design-by-analogy involves the access and transfer of similar characteristics or attributes from objects (ideas) in the source domain to objects (ideas) in the target domain to solve design problems in the target domain [59,60]. Based on design creativity theories, an analogical transfer can happen based on the surface or structural similarities [61]. In particular, computational and patent data mining methods have been proposed to aid in design-by-analogy. For example, Fu et al. developed a computational design support tool for identifying patent stimuli automatically at different analogical distances. By analyzing the textual information contained in the patent database, they built a function- and surface-Bayesian network, respectively [14]. The analogical distance between the design problem and stimuli is defined as the path length in the Bayesian network. Murphy et al. [13] proposed a functional vector method to represent patents into vectors to support the search of functional analogies across different vector distances. Song et al. [37] located patent stimuli in home, near, and far-fields with the patent technology network built on the patent bibliographic information. McCaffrey and Spector [12] developed a system named "Analogy Finder" to search adaptable stimuli in the patent database by rephrasing the input design problem into verbs and synonyms. Sarica et al. [41] used word embedding models to train vector representations of technical terms (which represent technical functions, components, structures,

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and working mechanisms) from the patent title and the abstract, to guide the retrieval of technical concepts according to their distances in the vector space.

One of the key factors influencing the effect of design-by-analogy is the analogical distance, which refers to the distance between the source and target fields [62]. The Conceptual Leap hypothesis states that far-field stimuli can provide the best inspiration for creative breakthroughs, owing to their surface dissimilarities [34–36]. Some anecdotal evidence exists in support of this statement. For instance, Chan et al. [62] observed that far-field stimuli led to new concepts of higher novelty but less quantity than near-field stimuli. In another study, Chan et al. [6] further observed that the most creative ideas are generated using near-field stimuli because of the easier perception and more obvious relevance to the original design problem. Srinivasan et al. [7] found that as the analogical distance increases, the novelty of newly generated concepts increases, but the quantity and quality decrease. Song et al. [37] also observed that near-field stimuli stimulate the most concepts, which exhibit a higher average novelty; combined home-field and far-field stimuli are beneficial to generate concepts in high quality. Goucher-Lambert and Cagan [8] and Goucher-Lambert et al. [9] found that semantically near stimuli increase the feasibility of design solutions, while distant stimuli improve their uniqueness. At the same time, owing to the experimental and empirical contexts, there also exist empirical findings that are inconsistent with those above [63]. Fu et al. [5] argue that stimuli from the "middle ground" (with moderate analogical distance) provide trade-offs leading to optimal benefits to design inspiration.

The use of visual or multimodal analogy for design inspiration has also been studied. Sio et al. [18] presented a meta-analysis study of industrial and engineering designers, which demonstrates the exposure to visual examples (sketch, photo, web page, etc.) produces the quality of the design output and measurable effects in novelty. Similarly, a study conducted by Goldschmidt and Smolkov [19] on the influence of visual stimuli has shown the benefits to industrial and engineering design students. Laing and Masoodian [20] reported positive benefits from the availability of visual examples to designers' experience during their ideation phase of the design process. Mckoy et al. [21] observed that the novice engineering designers with sketches of example designs create more novel and higher quality solution concepts on average comparing with text-based example designs. Linsey et al. [22] observed that the coupling of modalities (text and image) has significant potential to support innovation and enhance the design-by-analogy process. Sarkar and Chakrabarti [23] analyzed the effects of both nonverbal (image and video) and verbal (FBS, SAPPhIRE model) stimuli. They observed that nonverbal representations were more effective as stimuli than verbal ones. Their study also found that a nonverbal expression of stimuli can enhance the quality of the generated design concepts [23].

2.3 Patent Image Representation and Retrieval. A few researchers have proposed patent image retrieval methods, and most of them are SIFT based. These approaches often refer to specific descriptors. With a pre-trained vocabulary list, local features are projected into visual words and represent images in a format similar to documents, and classic indexing and searching schemes can be leveraged [64]. Huet et al. [26] proposed a relational skeleton method to capture the geometric features of a patent image based on the relationship between captured line segments. However, this system works well for some images while poorly for others. Also, the database used in this system is quite small. Vrochidis et al. [25] created the adaptive hierarchical density histogram (AHDH) to exploit the content of patent images. This method has been extended by Sidiropoulous et al. [65], who introduced the quantized relative density information. Both studies calculated the distribution of the binary pixels in a patent image by performing image segmentation. Csurka et al. [24] used Fisher vectors to represent patent images and then computed the pairwise similarity between the images based on their dot product.

In recent years, the convolutional neural network (CNN)-based methods have shown advantages over such SIFT-based models which only exploit simple geometric features within the images [32]. Features extracted from CNN have been shown to outperform geometric features in many computer vision tasks, such as visual object localization, image segmentation, and image classification. In the image retrieval field, some competitive results from the use of CNNs compared to traditional SIFT-based features have already been reported, even with low-dimensional vectors [66–69]. These CNN-based approaches usually extract compact representations from a specific fully connected layer in the CNN architecture. However, these prior studies only focused on general images rather than on technical images, such as patent images, which are usually colorless and intricate.

Taken together, prior research on CNN and its demonstrated advantages for representation learning of images have made it possible to develop a new patent image vectorization and retrieval method based on CNN features to support the design-by-analogy process. In the following section, we will present the methodology in detail.

3 Method

Our methodology includes training a novel convolutional neural network named Dual-VGG to derive design feature vectors that represent patent images. It measures their distances by cosine similarity in the vector space, to support design image retrieval for design-by-analogy. Specifically, we first present the Dual-VGG network architecture. The convolutional neural network aims at predicting the IPC section-label for each patent, which is equivalent to a classification task. In this study, we use the one-digit IPC label for each of the eight patent classes from A to H, while each class is comprised of smaller 3 to 7-digit subclasses hierarchically organized. Although one patent may belong to more than one class, the main class is chosen as its label. Thus, the study can be viewed as a multi-class single-label classification task. After the training process, we can extract features from the second-to-last layer of the network, which is a fully connected layer. Then, we can take all patent images in our database and represent each one as a specific vector. The dimension of the vectors can be fixed at a specific number, usually 512 or 1,024. In this study, we set 1,024 nodes for this fully connected layer. The cosine similarity method calculates the pairwise similarity between two images. In this way, we can find the nearest neighbors in the embedding space, which are also the most proximate patent images to a given image.

The framework for this research is shown in Fig. 1. The Dual-VGG network lies at the heart of the method. It enables the learning of rich image embeddings, containing not only the basic knowledge of shapes, colors, and patterns but also the technical-related knowledge contained in the IPC information. The quality of the features is mostly dependent on the performance of the trained model.

3.1 Dual-Visual Geometry Group Convolutional Neural Network. The Dual-VGG network is designed based on the VGG19 convolutional neural network [70], which was proposed by the Visual Geometry Group of the University of Oxford. The VGG19 is a robust convolutional neural network and is widely used in many applications related to computer vision. The VGG19 has 19 trainable layers, including convolutional layers, fully connected layers, max-pooling layers as well as dropout layers. The detailed architecture of the VGG19 refers to Ref. [70].

The Dual-VGG includes two VGG19 blocks, as shown in Fig. 2. The lower one is for visual material-type classification. Specifically, we add an auxiliary task to classify each patent image into a specific

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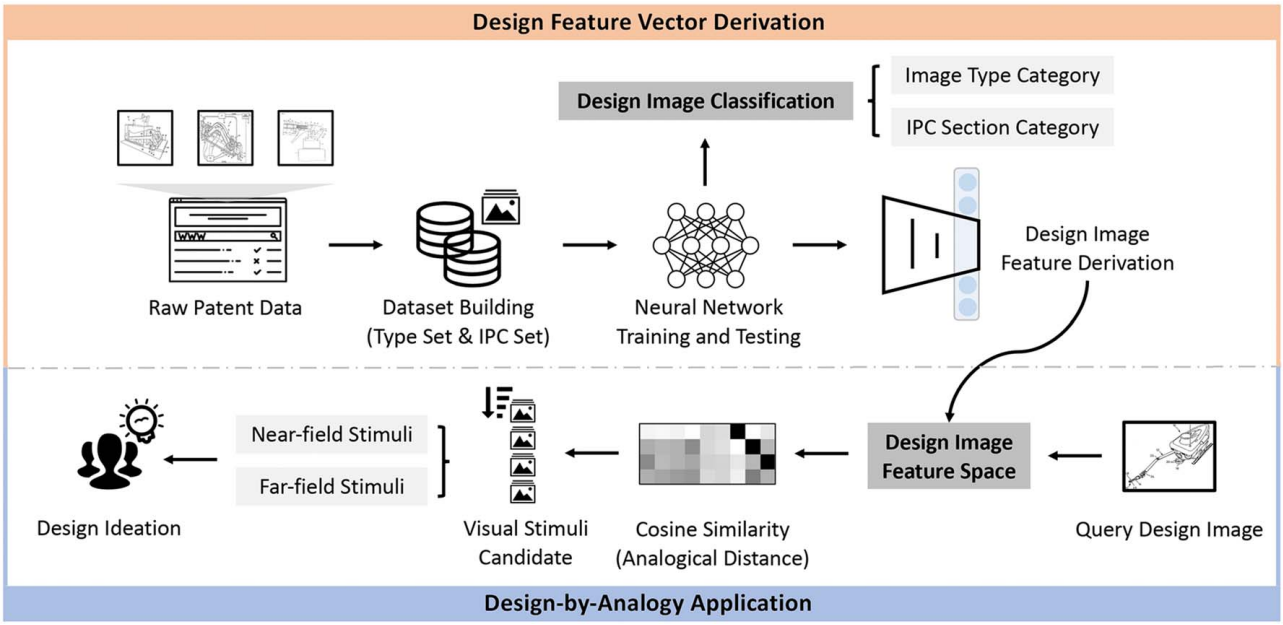


Fig. 1 The overall research framework of the proposed method

type group, which can also be viewed as a multi-class classification task. Vrochidis et al. claimed that there are nine types for patent images, including drawing, flowchart, graph, chemical structure, table, DNA, math formula, computer program, and symbol [25]. The purpose of the auxiliary task is to embed type-related knowledge into design features. During the training phase, we first train the lower block independently from the pre-trained weights. Then, we freeze all the weights in the lower block. The output vector from this block will embed visual characteristics for patent images. Meanwhile, the upper branch of the network aims to learn both the general geometrical knowledge and the technical-related knowledge contained in the IPC information.

In the network, we scaled the patent image into a $224 \times 224 \times 3$ vector for the input layer. There are several convolutional layers, activation layers, as well as max-pooling layers in both VGG19 blocks. The convolutional layers represent a convolution operation over the image using a feature map, and it is performed at each point; then, the result is passed to the next layer [71]. The filters belonging to the convolutional layers are some trainable feature extractors of usually a 3×3 size. Each convolutional layer stack is followed by an activation function and max-pooling operation. The activation function used here is a rectified linear unit (ReLU), which is a computationally efficient nonlinear function. After the activation function, a max-pooling layer is applied to our model. We take a filter of 2×2 size and a stride of the same length. In addition, the output value is the max number in every region that the filter convolves around [72]. In the fully connected layer, each neuron receives input from all the neurons in the last layer. A fully connected layer provides learning features from all the information of the previous layer, whereas a convolutional layer is usually based on consistent features with a small repetitive region. Through two VGG blocks, two 1,024-dimensional feature vectors are obtained.

The knowledge fusion strategy used in this research refers to intermediate fusion [73]. When all of the modalities are transformed into new representations, it becomes amenable to fuse different representations into a single hidden layer and then learn a joint multimodal representation. The majority of work in deep multimodal fusion adopts this strategy, where a shared representation layer is constructed by merging units with connections coming into this layer from multiple modality-specific paths. Following this strategy, the concatenation layer in our model connects two vectors into a 2,048-dimensional vector. Finally, after the concatenation layer, another fully connected layer takes all the features in the previous layer and generates the probability distribution for each IPC section label.

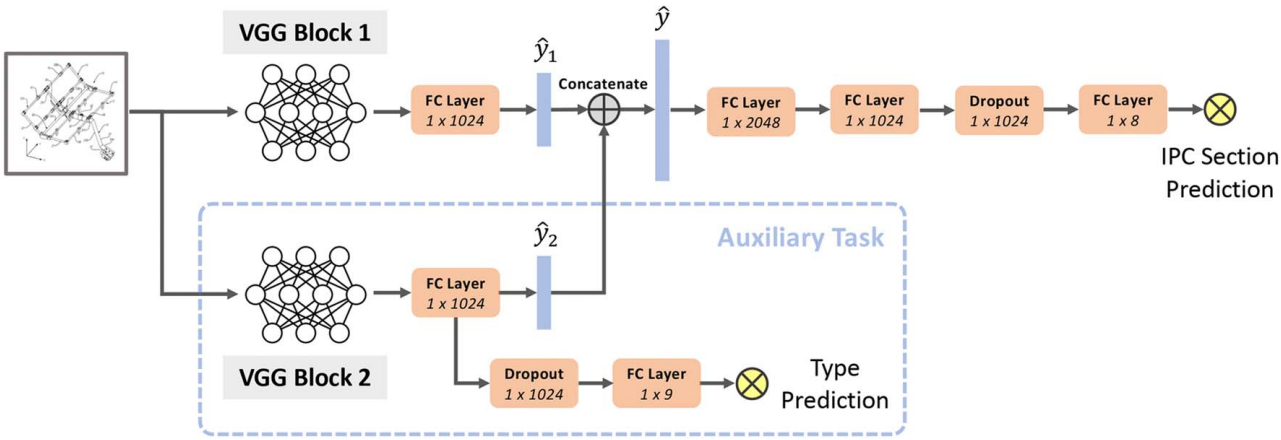


Fig. 2 The Dual-VGG network architecture

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Formally, in our model, let the patent image be I_i. We trained a specialized Dual-VGG network $f_v = I_i \rightarrow u_i$. For every patent image I_i, $i = 1$ to N, we can use our model to extract its features as a vector u_i. In this way, we converted the image space I into a latent feature space R_d. During the training process, the loss is calculated based on the categorical cross-entropy, and the back-propagation algorithm is used to change the weights and biases to make the network more predictive. The objective function is

$$L = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{K}(y_j^i \log(\hat{y}_j^i))$$

where N denotes the number of images in a batch and i is the index of an image. K denotes the number of classes, in this case, we have $K = 8$. y_j represents the output value of the last fully connected layer and we have $\sum_{j=1}^{K} y_j = 1$. $\hat{y}_j$ represents the ground truth; it equals 1 for the true class and 0 for others.

After the training process, the model can be used for both classification and feature deriving purposes. Since the goal of the model is to classify patent images into a specific IPC section, the design features extracted from the network will contain the technology-related knowledge learned from the IPC information.

3.2 Design Feature Deriving and Visual Stimuli Identification. Using the trained Dual-VGG network, we can transform the patent images into feature vectors. Specifically, each image is converted into a 1,024-dimensional vector from the second-to-last fully connected layer of the network. It is noteworthy that the number of nodes in the layer is a hyperparameter for the model, which can be modified as required. Additionally, our Dual-VGG network is capable of learning input data of different sizes, which means our learning model can be updated easily over time.

There are several metrics to measure the distance (or proximity) between two image vectors, such as Euclidean distance, cosine similarity, and Pearson's correlation. The cosine distance is one of the most easy-to-implement and efficient metrics, which is widely used in document and image retrieval tasks. Thus, based on the feature representations, we can compute the pairwise similarity between two feature vectors u_i and u_j using the cosine similarity

$$sim(u_i, u_j) = \frac{u_i^T u_j}{u_i u_j}$$

For this measure, the value of 1 indicates that the patent images are identical, and 0 (in the case of nonnegative 1,024-dimensional feature vectors) or less than 0 suggests that the images are completely dissimilar [74].

The vector space and the associations of vectors by cosine similarity enable at least two strategies of patent image retrieval for design-by-analogy. First, given a query patent image, we can compute its pairwise similarities to all other images and identify the most proximate images to it. This represents a direct image-to-image inference and search. Second, given a language-based design query, we can first use keywords to query for relevant patents using the USPTO website, Google Patents, Patsnap, Incopat, or other patent search tools, and then select the most relevant image(s) in the retrieved patent documents as the query image(s) to identify other images in our vectorized patent image database according to their cosine similarities. Both strategies can be used to discover visual design stimuli for design ideation and prior art searches. In both cases, design knowledge retrieval and discovery are conducted through the inference of design knowledge embedded in the images.

4 Experiment

4.1 Data Collection and Preprocessing. Our experiments are based on three data sets: a data set of image types used in the auxiliary task of the Dual-VGG network, An IPC section-label data set is used in the main task and a data set used for the case study that includes thousands of patent images from four technical domains. The distribution of images in the three data sets is reported in Fig. 3.[2]

Since the original patent database does not have the type labels for images, we utilized an existing patent image-type data set for the auxiliary task, the CLEF-IP 2011 [75], where all images have already been labeled into nine categories. To improve the final performance on the auxiliary classification task, the data augmentation technique was used to enrich our training data and obtain a balanced data set. Data augmentation methods such as random rotations or nonlinear deformations are effortless to implement and are effective in some settings at improving classification accuracy [76]. In this experiment, we used data augmentation to generate images that are shifted, zoomed in and out, rotated, flipped, distorted, or shaded with a hue from the randomly sampled images from each category. After data augmentation, we obtained 8,000 images for each category. The training set, validation set, and test set were separated as 6:1:1. Commonly used ratios range from 6:2:2 to 8:1:1. In this study, we tested 6:2:2, 6:1:1, and 8:1:1. The ratio 6:1:1 returned the best classification performance. Thus, we have 6,000 images, 1,000 images, and 1,000 images in the training, validating, and testing processes, respectively.

The IPC section-label data set is made up of all USPTO granted patent database from the years 2005 to 2009. The raw patent data were downloaded from the online website of the USPTO.[3] The bulk data include the full texts of patents since 1976 and the figures of patents in TIFF format since 2004. We collected 403,870 utility patents in eight categories at the IPC 1-digit level and their 10,877,766 images. Figure 4 shows illustrative examples of random patent images from eight IPC sections. It is noteworthy that more than one figure may be shown on the same page (a compound figure). In this case, we used a fine-tuned YOLO-V3 neural network [77] to detect the figures and segment them. Finally, a total of 13,998,254 figures were obtained.

To evaluate the performance of the representation learning and make sense of the derived vector representations and distances of patent images, we curated an evaluation data set that contains patents in four different technological domains related to mechanical design: flywheel, milling machine, aircraft transport, and robotics. The classification overlap method (COM), developed by Benson and Magee [78,79], was used to obtain a relatively relevant and complete patent set of a certain specific domain. The COM had been successfully applied in several prior studies [80,81]. The relevancy rate of the patents in each domain is reported in Table 1 [79,82]. Note that some irrelevant patents involved in this data set can also be viewed as far-field stimuli for designers in design-by-analogy process. About 15,851 patents were obtained from these four domain sets. Then, we downloaded all the figures of these patents from the PATSNAP[4] website. Only images on the front page were used in this data set with the assumption that they are more important and representative of the whole patent than the other figures.

4.2 Neural Network Training. The neural network architecture and classification process has been implemented in PYTHON using the Keras library,[5] which provides an application interface for loading pre-trained models. We first trained half part of the model designated individually to the auxiliary task, and we froze

[2]In this figure, "CH" represents "Chemistry," "DR" represents "Drawing," "FL" represents "Flowchart," "GR" represents "Graph," "MATH" represents "Math formula," "PRO" represents "Programming," "SYM" represents "Symbol," "TAB" represents "Table"; "AIR" represents "Aircraft," "FLY" represents "Flywheel", "MM" represents "Milling Machine," and "ROBO" represents "Robotics."

[3]https://bulkdata.uspto.gov

[4]https://www.patsnap.com

[5]https://github.com/keras-team/keras

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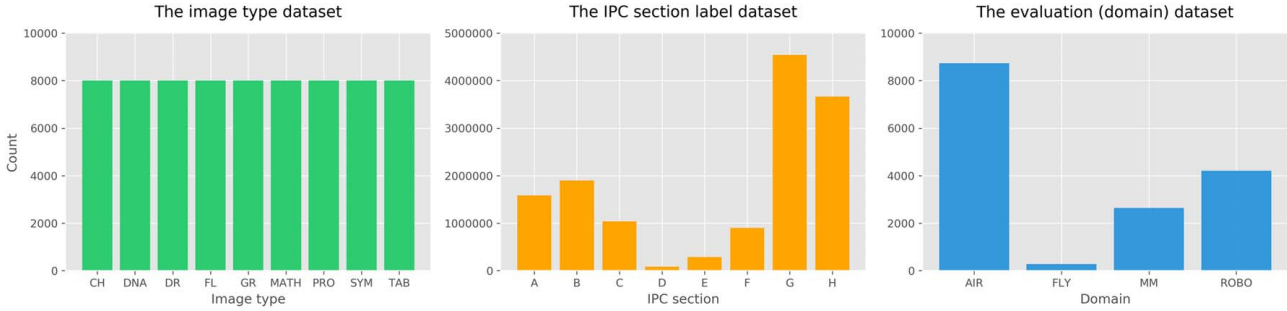


Fig. 3 The distribution of three data sets

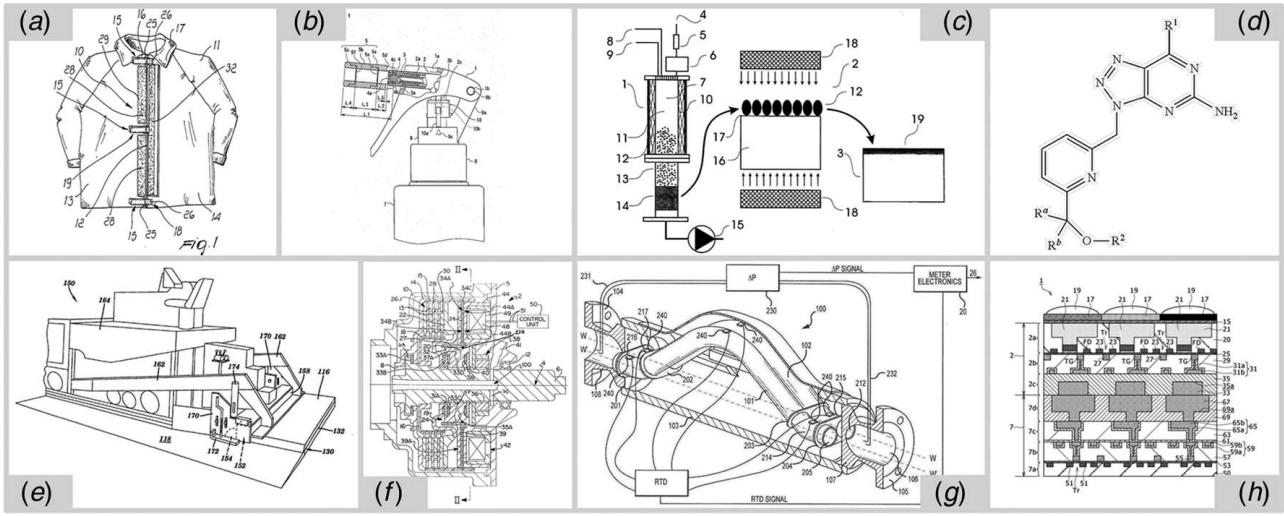


Fig. 4 Examples of random patent images from eight IPC sections: (*a*) human necessities, (*b*) performing operations, (*c*) chemistry metallurgy, (*d*) textiles and paper, (*e*) fixed constructions, (*f*) mechanical engineering, (*g*) physics, and (*h*) electricity

all the weights after training. Then, we trained the rest of the model. During the training process, the data set performed forward and backward passes to update the parameters of the model, including both the weights and bias. When the forward and backward passes finish passing the whole data set, it is called one epoch. For other hyperparameters, we set 1E-4 for the learning rate of the SGD optimizer and 32 for the batch size.

The network was trained for 50 iterations for the auxiliary task and 10 iterations for the IPC section-label prediction task. The training process for each task took 16.8 h and 136.2 h, respectively. We employed the 4 GPUs (2 Titan Xp 16GB and 2 Titan N 16GB) with the Intel core i7 processor to train deep learning models more rapidly.

4.3 Training Results. In this work, we trained the Dual-VGG neural network for classifying patent images into different visual material types (called an auxiliary task below) and into explicit IPC 1-digit label categories (called the main task below). For a benchmarking and evaluation purpose, we also implemented the same data sets and tests, using ResNet50 [83], DenseNet121 [84], as well as XRCE's method [24].

ResNet50 and DenseNet121 are the most popular and successful convolutional neural networks for image classification tasks. XRCE's method was the first-place winner of the image-type classification task in the CLEF-IP 2011 contest, which achieved a 92.2% accuracy. XRCE's method represented images with Fisher Vectors and trained many one-versus-all linear classifiers. Table 2 shows the accuracy of the test data of each model. Dual-VGG outperforms all other models in both auxiliary and main tasks. Three CNN-based methods are better than XRCE's method in the type of classification task. This is because convolutional neural networks can learn richer features of images due to its deep architecture compared to the classical local descriptors. Thus, CNN-based retrieval

Table 1 The relevancy rate of the patents in four domains

Technological domain	Size	Relevancy	IPC subclass (four-digit code)
Aircraft	8736	79%	B64D; B64C
Flywheel	274	70%	F16F; H02K
Milling machine	2636	93%	B23C
Robotics	4205	93%	B25J

Table 2 Classification accuracy of Dual-VGG, ResNet50, DenseNet121, and XRCE's method on the test set

Method	Accuracy	
	Type prediction	IPC section prediction
Dual-VGG	**95.37%**	**54.32%**
ResNet50	92.54%	49.67%
DenseNet121	94.69%	52.45%
XRCE's	92.20%	/

Note: The bold-faced values (from the proposed method) represent the best performance among all methods.

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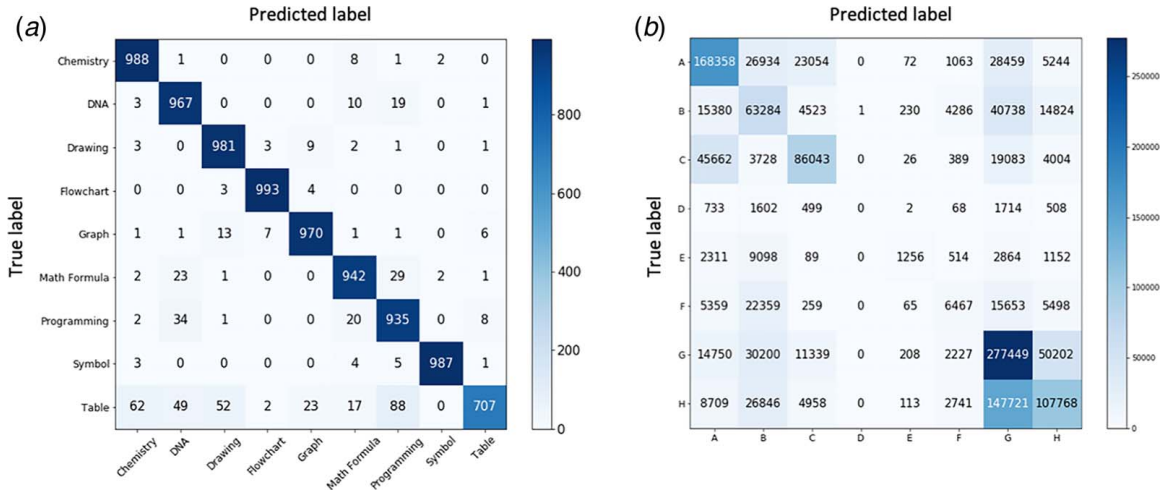


Fig. 5 (*a*) Confusion matrix of the auxiliary task and (*b*) confusion matrix of the main task

methods have been continuously proposed in recent years and are gradually replacing the handcrafted local detectors and descriptors [32]. For the main task, Dual-VGG performed better than the other two neural networks, showing the highest accuracy of 54.32%. This is because the Dual-VGG network combines the auxiliary task and main task in the same model architecture, while ResNet50 and DenseNet121 implement two tasks independently. It shows that involving type information in the model is useful to predict the IPC sections. However, we note that accuracy (54.32%) is not that high. We believe there are two reasons. One reason is that it is naturally difficult to predict the IPC section-label only with single patent images even for human experts. For example, in some figures, only the partial view of a specific design is shown, which cannot represent the full design information of the patent. Another is that the resolutions of some patent images are low.

Figure 5 shows the confusion matrix of the Dual-VGG network. From Fig. 5(*a*), it is clear that our model performs well for nearly all visual material types except tables. Figure 5(*b*) shows our model learned more information from IPC sections A, B, C, F, G, and H compared to sections D and E. This is because that the data set used for training is unbalanced. We can observe that the model ignores class D as this group is quite small. The two confusion matrices show that our model learned specific knowledge about types and the IPC information. Figure 6 illustrates a sample subset of images in the trained vector space. All the images in this map are sampled from our evaluation data set. The features derived from the trained Dual-VGG model are projected into a two-dimensional space using the *t*-distributed Stochastic Neighbor Embedding (t-SNE) method [85].

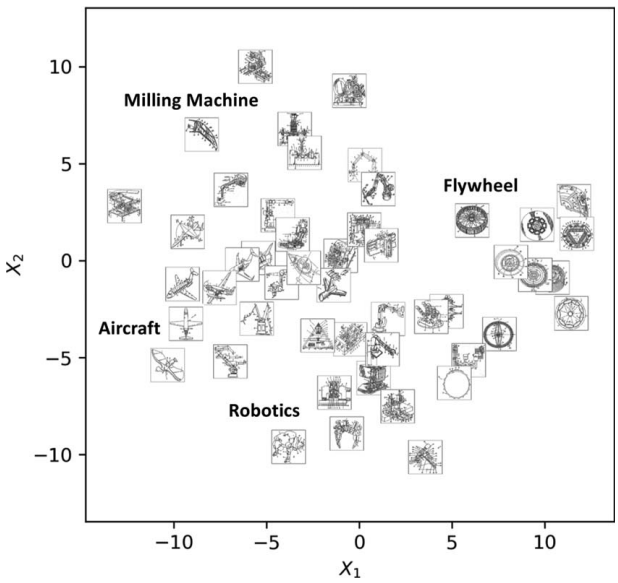


Fig. 6 Illustration of a sample set of images in the trained vector space

5 Evaluation Case Study

Presently, robots are increasingly used in various working environments to replace manual work, especially to perform repetitive tasks. The robot arm is a kind of robotic manipulator, usually programmable, with similar functions to the real human arm. The manipulators' links are usually connected by joints, which allow rotational and translational motion. Those links can be regarded as a kinematic chain. The end of a kinematic chain in the manipulator is called the end effectors and is similar to human hands. The end effectors are usually designed to perform the kinds of tasks that depend on applications requiring high accuracy, such as welding, grasping, and spinning. The robot arms can be manipulated manually or controlled automatically for industrial or home applications [86]. The history of robot arms dates back to 1959 in the US when George Devol designed the first robotic arm named "Unimate" to lift and stack hot metal parts [87]. Since then, many industrial companies and academic institutions have made attempts to develop robot arms, such as KUKA, ABB, Hitachi, and Boston Dynamics.

Figure 7 shows the ten images most related to the keyword "robot arm" retrieved via a keyword search in Google Images. We can observe that these Google-retrieved images represent merchandise information from online shopping platforms or robot companies, such as Amazon, Ufactory, and Ozrobotics. The appearance and functions of these robot arms are highly similar in appearance. Such retrievals do not provide broad design concepts nor sufficient engineering design information about robot arms to stimulate design ideation.

For comparison, we conducted a keyword-based search in the USPTO patent database using PATSNAP and extracted patent images from these patents. Table 3 lists the top 10 relevant patents. Figure 8 shows the top 10 images extracted from the first page of these patents. It is evident that these returned patents are highly relevant as most of their titles include the query keyword "robot arm." However, the design information contained in some of these patent images is difficult to absorb. For example, Image Keyword-2 provides a flowchart about a certain kind of robot

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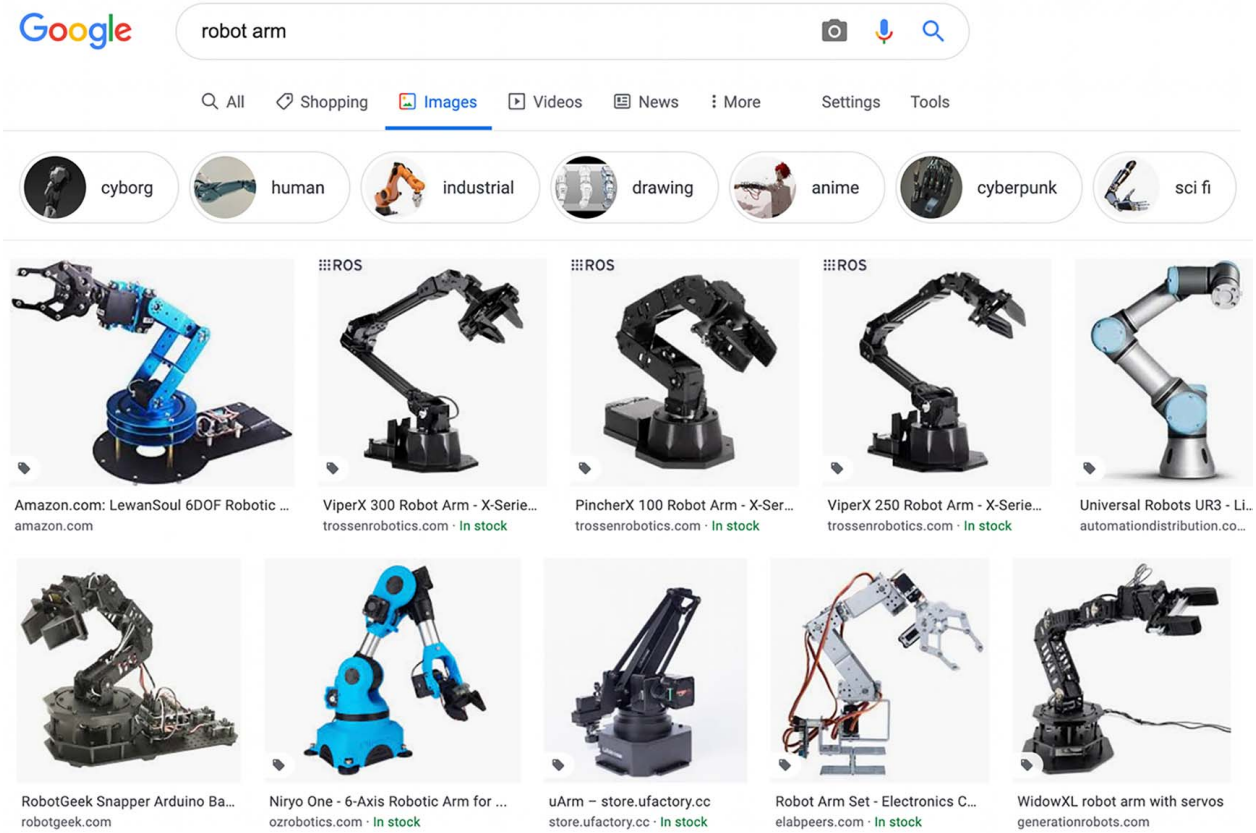


Fig. 7 Most relevant images to "robot arm" as retrieved from Google Image

control system, and Keyword-5, 8, and 10 only use simple graphics to explain the working principles of the specific design. To fully understand the underlying ideas, designers would have to read the patents' full texts, which is tedious and time-consuming. By contrast, it would be preferable if the retrieved images could directly inspire engineers with their visual information.

Based on the proposed method in this paper and the trained Dual-VGG network, we have derived the 1,024-dimensional vectors for all the patent images in the database as described in Sec. 4.1. To retrieve robot arm design stimuli, we started with an initial query image, which is the first image in Fig. 9 and describes a coupler for connecting a robot arm of a remote manipulator device to a tool body, and retrieved other images in the embedding space according to their pairwise similarities with the query image.

Table 3 Most relevant patents to "robot arm" as retrieved from Patsnap using keyword searching

	Patent ID	Title
1	US10005182	Robot arm
2	US4920500	Method and robot installation for programmed control of a working tool
3	US4636138	Industrial robot
4	US6430472	Robot feature tracking devices and methods
5	US8290621	Control apparatus and control method for robot arm…
6	US8185243	Robot, control device for robot arm and control program for robot arm
7	US8909374	Robot arm control apparatus, robot arm control method, robot…
8	US7040852	Robot arm mechanism and robot apparatus
9	US10471596	Robot arm and unmanned aerial vehicle equipped with the robot arm
10	US8340820	Robot arm and method of controlling robot arm to avoid collisions

Note that the initial query image used in the experiment was selected by authors considering both the visual and semantic content of the corresponding patent. Designers may choose different initial query images for their preferences. Figure 9 shows the top nine most proximate images to the query image via vector space inferences. Table 4 reports the patent ID and titles of the patents that contain these images.

In Fig. 9, DVGG-2, DVGG-4, and DVGG-5 are in the traditional form of robot arms, which are comprised of motors to drive the wrist mechanisms. DVGG-1 presents a novel end-effector with high flexibility, which is made up of several articulated booms. DVGG-3 shows a motion conversion device used in the robotic joints of robot arms. DVGG-6 presents a moveable linkage for a parallel mechanism used in the mechanism design process of the robot arms. DVGG-9 shows a mechanical mechanism. These design images show design configurations and mechanisms that can be used in or together with robotic arm designs. Furthermore, DVGG-7 and DVGG-8 show a pneumatic gripper and a grapple fixture with improved docking and engaging capabilities as an irregular design. These images describe designs that are not about typical robot arm designs but can be potentially related to robot arms for new designs.

Compared to the images in Fig. 8 identified from the patent documents retrieved via a keyword search, the patent images in Fig. 9 cover design concepts that are broader than robot arms specifically but can be easily related to robot arm designs. It is important to note that these images are retrieved according to their design feature vector similarities with the query image in the design feature space derived from the Dual-VGG network training. Thus, retrieving highly proximate designs in the design feature space may discover images that are not directly related to the query on the surface, e.g., DVGG-7 and DVGG-8 in Fig. 9.

Such design retrievals are particularly valuable for augmenting design ideation and supporting design-by-analogy. Although some of the returned patents in Table 4 do not include "robot

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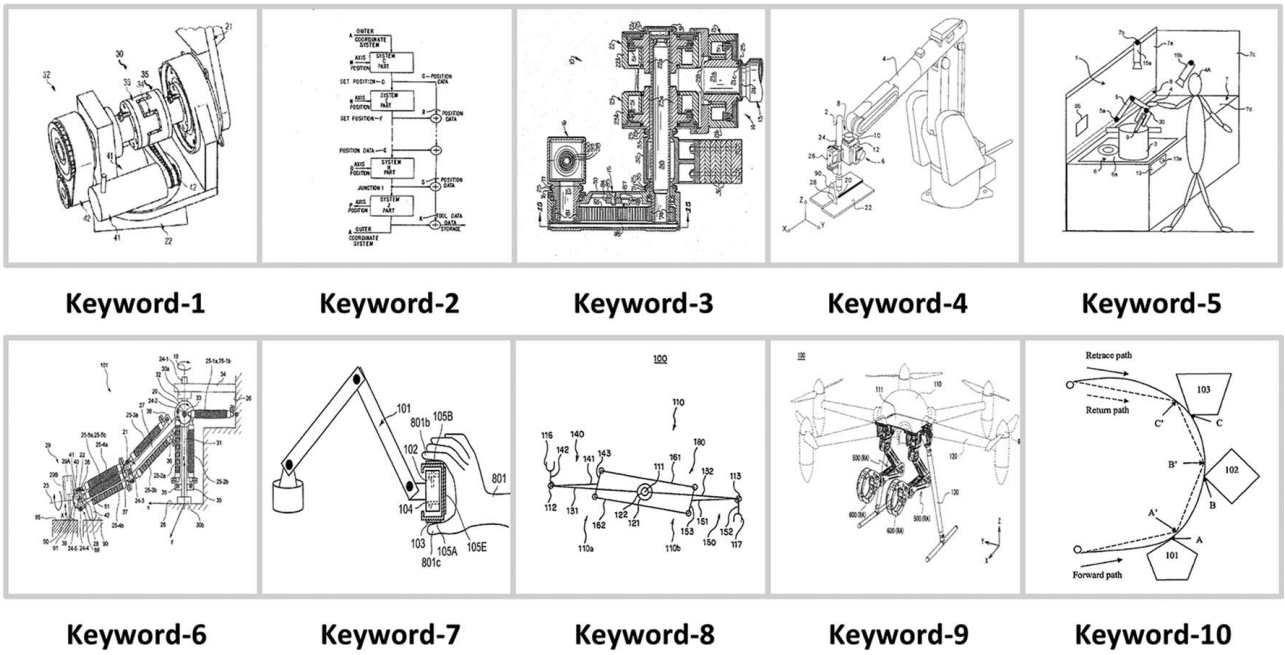


Fig. 8 The patent images on the first pages of the patents reported in Table 3 (retrieved by keyword searching)

arm" term in their titles, nearly all of them can be easily related to the given topic from a mechanical design perspective. These patents in Table 4 are identified via a visual search, in contrast to those patents in Table 3, which were identified via a text search.

To compare the proposed image retrieval method with other CNN-based methods, we conducted the same retrieval task using the Resnet50 with pre-trained weights from ImageNet. Similarly, we derived the design feature vectors from the second-to-last layer of the model and then used the same query image to identify the nine most relevant patent images, shown in Fig. 10. Compared with Fig. 9, one can easily observe that the retrieved images using Resnet 50 present a shape highly similar to the query image, especially the images RES-1, 3, 4, 5, and 8. Such retrievals may result from that the pre-trained ResNet50 considers only the visual characteristics but not the technology-related information.

Moreover, we experimented with another indirect design retrieval strategy. As described earlier, we obtained ten relevant patents and their images using a keyword search (Fig. 8). Then, we manually read them to identify four visually valuable items as the query images for inferences in the design feature vector space to identify other images in the database. Figure 11 shows the images in our database, which are most proximate to the four figures in the retrieved patent set. Compared to those in Fig. 8, these images show more visually diverse and informative design information. With the hybrid keyword-to-image retrieval method and single query image-driven retrieval method, the design

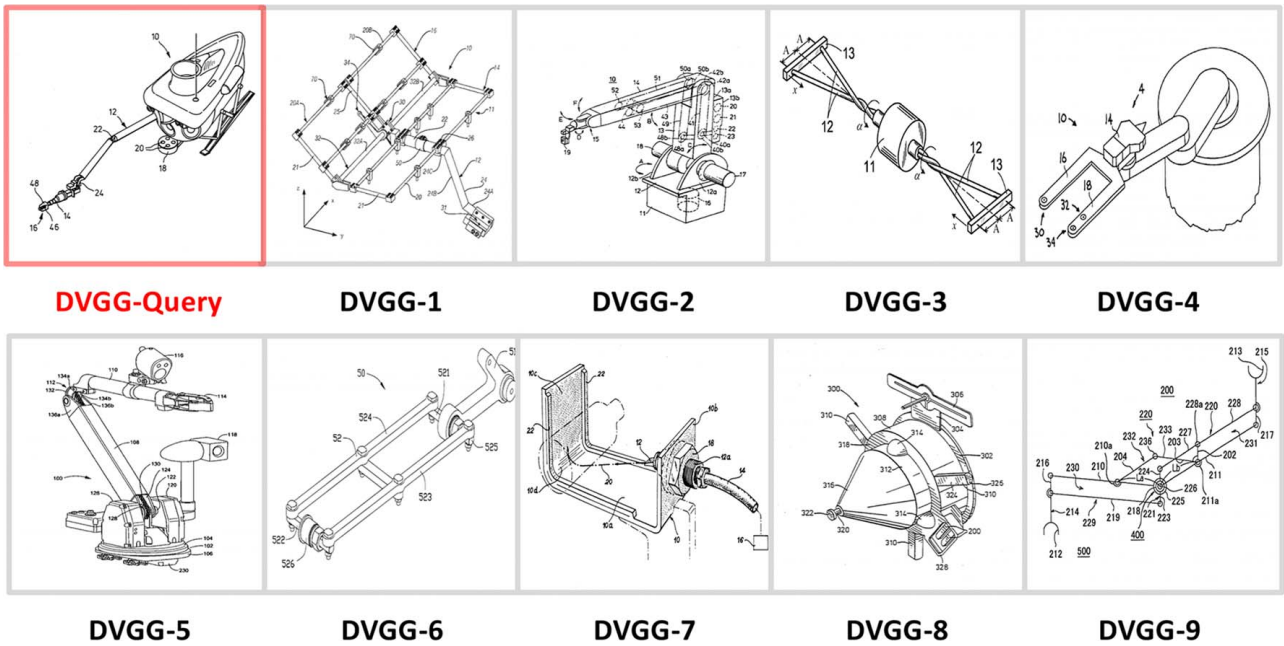


Fig. 9 Top nine most proximate patent images to the given query image in the database as retrieved by the trained Dual-VGG model

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Table 4 The ID and titles of the patents containing the patent images in Fig. 9

	Patent ID	Title
Query	US4227853	Manipulator wrist tool interface
1	US5467889	Nestable elastic fuel tank and method for making same
2	US4251040	Wind driven apparatus for power generation
3	US6623235	Robot arm edge gripping device for handling substrates…
4	US7571526	Multi-blade router tool, edger with multi-blade router tool…
5	US6837462	Boom load alleviation using visual means
6	US8844860	Foldable rise and stare vehicle
7	US6231002	System and method for defending a vehicle
8	US8777165	Aircraft fuel system
9	US6116845	Apparatus for supporting a workpiece for transfer

knowledge discovery and design opportunity exploration are augmented through the inference of the design knowledge contained in the related images.

It is noteworthy that the foregoing case examples focused on the most proximate images to the query for demonstrative purposes. Although the examples are sufficient to reveal that our such images as DVGG-6, DVGG-7 and DVGG-8 retrieved from our model diverge from those retrieved through keyword-based and pure-visual search and come beyond the specific robot arms, they are nonetheless among the most proximate ones to the query image and are generally near-field stimuli based in our design feature vector space. In practice, for those designers interested in design novelty and far-field stimuli, they may explore more distant patent images based on the cosine distances in the trained design feature vector space. The design creativity literature [5–11,34–37] has suggested near stimuli are beneficial for effectiveness and quality of design ideation, whereas far stimuli give rise to idea novelty and chances for breakthrough innovation. The cosine distance information between the CNN-based design feature vectors of different images provide a new basis for the inferences, searches, and retrievals of near-to-far visual stimuli from the patent database to support design-by-analogy. With such guidance, designers may explore and choose patent images of varied cosine distances to a given design problem in the design feature vector space trained with our model, for their varied interests in design inspirations for effectiveness and quality versus novelty and radical innovation opportunity.

6 Concluding Remarks

In this paper, a CNN-based approach is developed to derive design feature vectors for patent images to support visual design stimuli retrieval for design-by-analogy. The core of this method is to represent patent images embedding not only superficial visual information but also technology-related knowledge. The representation learning process is conducted through the model training of two different tasks: visual material-type prediction and IPC section-label prediction. This research contributes to the growing studies about data-driven design, analogy-based design, and machine learning-based design analytics.

In practice, users can use our CNN-based method to pre-derive the design feature vectors of the patent images in a patent database of their own choice and store the trained vectors. Then, the pre-derived vectors can be directly applied to support the inferences and retrieval of patent images that have varied vector distances from a given query image within the database. When the images that represent the design stimuli or query of interests are outside the ones with pre-trained vectors, one can apply the neural network with trained weights to generate design feature vectors of the new images first. Figure 12 depicts the workflows for different kinds of queries. Furthermore, when designers have multiple queries, they can derive the design feature vectors for all query images and combine them into one vector (e.g., through averaging operation) for further use. Designers can also use the previously retrieved images as the queries for subsequent searches to iteratively identify more stimuli.

This work still has several limitations. First of all, we found that not every patent image can be used to represent the information of the whole patent. Some patent images are not in high quality. The retrieval results are related to the given query image to some extent. Second, alternative network parameters in the training might potentially lead to better results. In addition, while very

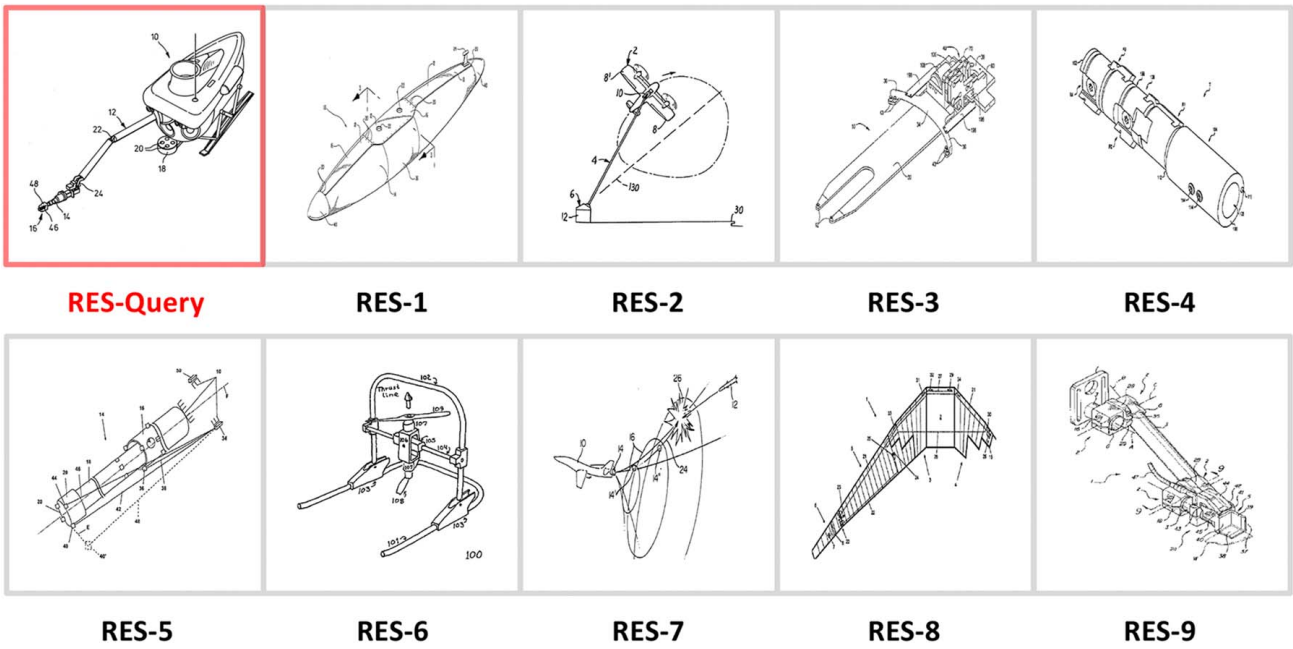


Fig. 10 Top nine most proximate patent images for the given query image in the database as retrieved by the pre-trained ResNet50 model

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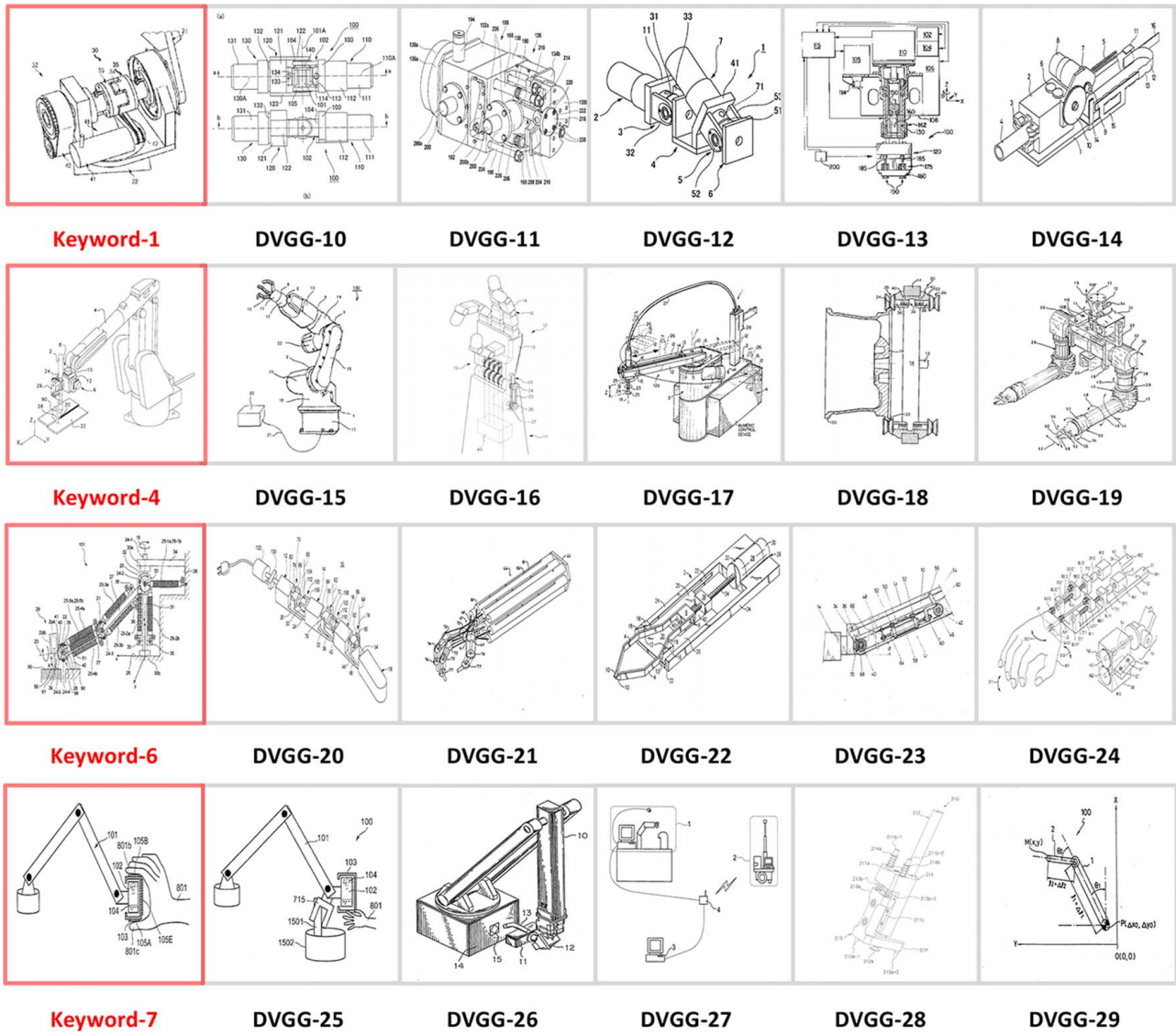


Fig. 11 Patent images retrieved from the four query images

often images can be self-explanatory and specific enough to provide inspiration, sometimes it might be difficult for designers to make sense of an image alone. In such cases, adding related text content to the image might aid the sense making of the image. Furthermore, this paper presents only a single small-scale case study. To further validate the proposed method, more diverse contexts, more systematic tests, and more human evaluators are desirable.

In future work, on the one hand, we will build a more complete patent image database related to mechanical engineering domains and use the proposed method to derive a feature vector for every patent image for the wider uses of other interested researchers. On the other hand, we will continue to improve the neural network model and try to develop a more robust architecture to learn better design features from patent images. Finally, we plan

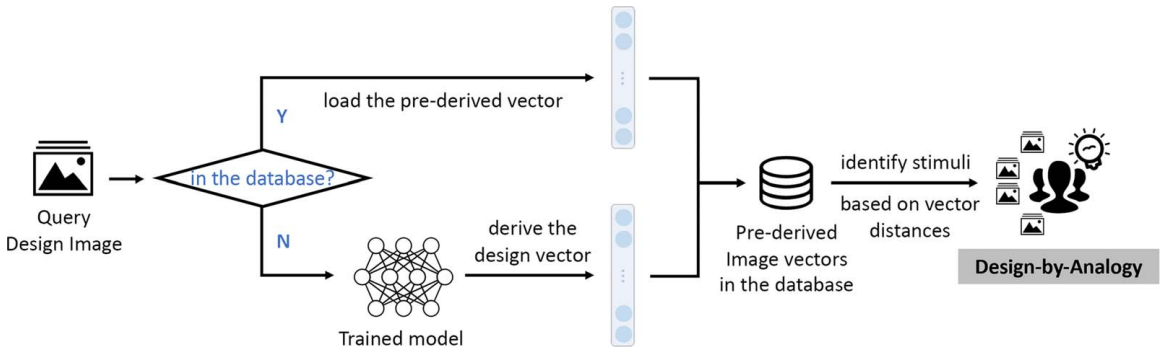


Fig. 12 The workflow for different kinds of queries

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to conduct a comprehensive human study to develop more nuanced understanding of the ideation performance outcomes from using design stimuli of varied distances as determined by the feature vectors from the proposed method.

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